

SUPPLEMENTARY MATERIAL FOR SMN-ATTACK: SALIENT REGION-AWARE MULTI-SCALE NEIGHBORHOOD ATTACK FOR TRANSFERABLE VISION-LANGUAGE MODELS

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A. DETAILS OF THE MOTIVATION STUDY

This section gives the detailed settings for Table 1 in the main paper.

A.1. Uniform Average-Budget Setting

Let $M_0 \in \{0, 1\}^{H \times W}$ denote the saliency mask extracted from the original image. The salient-region area ratio is

$$R_{\text{area}} = \frac{\|M_0\|_1}{HW}. \quad (\text{A1})$$

For the region-wise setting, the perturbation budgets are ϵ_s and ϵ_b for salient and non-salient regions, respectively. The corresponding average budget is

$$\epsilon_{\text{avg}} = R_{\text{area}}\epsilon_s + (1 - R_{\text{area}})\epsilon_b. \quad (\text{A2})$$

In the uniform average-budget setting, all pixels use the same budget ϵ_{avg} :

$$\|I^{\text{adv}} - I\| \leq \epsilon_{\text{avg}}. \quad (\text{A3})$$

In the region-wise setting, the constraint is

$$\|I^{\text{adv}} - I\| \leq \epsilon_s M_0 + \epsilon_b(1 - M_0), \quad (\text{A4})$$

where M_0 is broadcast to image channels when necessary. Thus, the two settings have the same average perturbation budget but different spatial budget distributions.

A.2. Region Sensitivity Metrics

We report R_{area} , R_{grad} , and Γ in Table 1. The area ratio R_{area} is defined above. The metric R_{grad} measures the proportion of effective gradients located in salient regions.

For an attack variant a , let I_t^a be the adversarial image at iteration t , and let ϵ^a denote its perturbation bound. For the uniform average-budget setting, $\epsilon^a = \epsilon_{\text{avg}}$. For the region-wise setting,

$$\epsilon^a = \epsilon_s M_0 + \epsilon_b(1 - M_0). \quad (\text{A5})$$

We first remove pixels that have reached their perturbation bounds:

$$E_t^a = \mathbf{1}[\|I_t^a - I\| < \epsilon^a]. \quad (\text{A6})$$

A gradient is defined as non-saturated if it can push the pixel further or pull it back into the interior. Specifically, letting $g_t^a = \nabla_{I_t^a} L_{\text{img}}(I_t^a, T^*)$, the effective gradient indicator E_t^a is defined as:

$$E_t^a = 1 - \mathbf{1}[(I_t^a \geq U^a, g_t^a > 0) \text{ or } (I_t^a \leq L^a, g_t^a < 0)]. \quad (\text{A7})$$

The effective normalized gradient is

$$\hat{g}_t^a = E_t^a \odot \left| \frac{g_t^a}{\|g_t^a\|_1} \right|. \quad (\text{A8})$$

Then R_{grad} is computed as

$$R_{\text{grad}}^a = \frac{\|M_0 \odot \sum_{t=0}^{N-1} \hat{g}_t^a\|_1}{\|\sum_{t=0}^{N-1} \hat{g}_t^a\|_1}. \quad (\text{A9})$$

Finally, the relative effective gradient density is

$$\Gamma^a = \frac{R_{\text{grad}}^a}{R_{\text{area}}}. \quad (\text{A10})$$

A value of $\Gamma > 1$ indicates that salient regions receive a larger effective gradient proportion than their area proportion.

A.3. Protocol

We apply the region-wise allocation strategy to SGA and SA-AET, while keeping their remaining components unchanged. For each attack, we compare the uniform average-budget setting with the region-wise setting. The same mask M_0 is used to compute R_{area} , R_{grad} , and Γ in both settings.

Attack performance is measured by black-box attack success rate on Rank-1 text-to-image retrieval and image-to-text retrieval, denoted as TR ASR and IR ASR. As reported in Table 1 of the main paper, region-wise allocation increases both the effective gradient concentration and the black-box ASR under the same average perturbation budget.

B. ADDITIONAL IMPLEMENTATION DETAILS

For SMN-Attack, we run each main experiment three times with different random seeds and report the median and range (max – min).

B.1. Image Attack

DINO ViT-S/16 is used to extract CLS-to-patch attention maps. The attention map is upsampled to the image resolution and normalized before thresholding. We remove small isolated connected components and preserve the two largest connected components. The smoothed mask is initialized as $\bar{M}_0 = M_0$ and updated with $\beta = 0.6$.

For random multi-scale neighborhood aggregation, we use $K = 3$ scales with weights $\{0.5, 0.75, 1.25\}$. The noise scale is computed as $\sigma_k = \sigma_0 w_k / w_0$, where $\sigma_0 = 0.01$ and $w_0 = 0.5$. Random crops use ratios in $(0.8, 1.0)$ and are resized back to the original resolution.

B.2. Text Attack

For adversarial text generation, we set the token substitution budget to $\epsilon_t = 1$. Candidate tokens are generated by a masked language model. We keep the top 10 candidates for each token position and discard candidates with prediction probability lower than 0.3. Candidates are evaluated by Eq. (15) in the main paper, and the best-ranked substitution is applied to obtain T^{adv} .